

Quick Check

Lesson 7: Microscope Practical on Mitosis: Catching Growth in Action in Onion Root Tips

8 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.4: Cell Division | Lesson 7

A onion (kitunguu) root grown in water for a week lengthened by 3 cm. Which part of the root is making most of the new cells that make it longer?

- A** The tip region, where cells are dividing
- B** Nowhere, because the existing cells just get bigger
- C** The middle only, where the root is oldest
- D** The whole root equally, from the tip to the top

In preparing an onion root tip squash, the tip is warmed in 1M hydrochloric acid for a few minutes. Why is this done?

- A** To soften the tissue so the cells can be squashed into a thin single layer
- B** To colour the chromosomes
- C** To make the cells divide faster
- D** To make each cell grow bigger

Under the microscope, a learner sees a cell in an onion root tip where the dark-stained chromosomes are all lined up across the middle (equator) of the cell. Which stage of mitosis is this?

A Metaphase

B Telophase

C Prophase

D Anaphase

A group counts 80 cells from an onion root tip squash. Of these, 60 are in interphase and 20 are in the stages of mitosis (prophase, metaphase, anaphase or telophase). What percentage of the cells are in interphase?

A 20%

B 75%

C 25%

D 60%

A learner is looking at two cells in an onion root tip. What clue tells you that a cell is in anaphase and not metaphase?

- A** Chromosomes are lined up along the middle of the cell
- B** Sister chromatids are being pulled apart towards opposite ends of the cell
- C** Two separate new nuclei have formed and the cell is dividing in two
- D** Chromosomes are thin and thread-like inside an intact nucleus

Learners see many cells at different stages of mitosis in an onion root tip, while cells further from the tip are not dividing. Which statement is best supported by this evidence?

- A** New cells are made when existing cells in the tip divide, and each new cell receives a copy of the chromosomes
- B** The root grows only because its cells get bigger
- C** Most cells in the tip are dividing at exactly the same moment
- D** Food turns directly into new cells at the tip

One cell divides by mitosis to make two cells, and every cell divides again every 24 hours. Starting with 1 cell, how many cells will there be after 10 days?

A 100

B 512

C 20

D 1024

A Kenyan child who eats well grows more than a child who does not get enough food. Which explanation fits what we learned about mitosis?

- A** Food makes cells divide without needing chromosomes
- B** Food turns directly into new cells in the body
- C** Food makes each cell grow bigger so no cell division is needed
- D** Food gives the materials and energy that cells need to divide and make new cells

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).